

THE ECONOMICS OF GENDER

Recognition for Group Work: Gender Differences in Academia[†]

By HEATHER SARSONS*

Do promotion gaps between demographic groups emerge when employees work in groups and employers cannot perfectly observe employee effort or ability? Organizations increasingly rely on group work for production. Lazear and Shaw (2007) report that the share of large US firms that predominantly rely on group work for production rose from 27 percent to 78 percent between 1987 and 1996. Yet there is little empirical evidence on how credit for group work is allocated and whether demographic characteristics such as gender play a role.

In this paper, I test whether uncertainty over an individual's role on a project leads to differential attribution of credit that contributes to the gender promotion gap. In many industries, women are not only hired at lower rates than men are, they are also promoted at lower rates. A significant portion of the promotion gap remains unexplained even after accounting for factors such as productivity, personality and behavioral differences, and fertility preferences.¹

I look at tenure decisions within academia, focusing mainly on economics, to answer this question. Economics is an ideal testing ground as (i) there is a large tenure gap between men and women that has not shifted despite more women entering the field, and (ii) authors are

ordered alphabetically on papers, making it difficult to discern an individual's contributions in a coauthored paper. Solo-authored papers, on the other hand, provide a relatively clear signal of a person's skill. Correlating the fraction of an individual's papers that are coauthored with tenure, I test whether women receive less credit for work when their contribution is unclear and whether the tenure gap shrinks when women solo-author (i.e., send clear signals).

Overall, I find that men and women who solo-author most of their work have similar tenure rates conditional on the quality of papers. However, an additional coauthored paper is correlated with an 8 percentage point increase in tenure probability for men but only a 2 percentage point increase for women. This gap is significantly less pronounced for women who coauthor with women, suggesting that the attribution of credit is related to the gender mix of coauthors. I contrast these results with sociology, a discipline in which coauthors are listed in order of contribution. When contributions are made clear, men and women benefit equally from joint work.

I. Data

To look at the relationship between paper production and tenure, I construct a dataset using CVs from economists who went up for tenure between 1985 and 2014 at top 30 PhD-granting universities² in the United States. To account for people who went up for tenure, were denied it, and moved into industry, non-US schools, or non-top-30 schools, I collect historical faculty lists from 16 of the 30 schools and locate over 90 percent of faculty who ever went up for tenure at these 16 institutions. I also look through

*Harvard University, Littauer Center, 1805 Cambridge Street, Cambridge, MA 02138 (e-mail: sarsons@fas.harvard.edu).

[†]Go to <https://doi.org/10.1257/aer.p.20171126> to visit the article page for additional materials and author disclosure statement.

¹There is a large literature documenting gender differences in productivity, attitudes toward different types of work, and family choices. See, for example, Niederle and Vesterlund (2007); Antecol, Bedard, and Stearns (2016); Ceci et al. (2014); and Ginther and Kahn (2004).

²Ranking is from <https://ideas.repec.org/top/top.usa.html>.

CVs of individuals at the top 75 US institutions, the top 5 Canadian institutions, and the top 5 European institutions to locate anyone who went up for tenure at a top 30 US school and then moved to another school. Finally, I check economists' CVs at the major Federal Reserve Banks in the United States.

From individuals' CVs, I code where and when they received their PhDs, their employment and publication history, and their primary field. To determine whether someone received tenure, I follow the guidelines on each school's website as to when tenure decisions are made. The majority of schools require faculty to apply for tenure after seven years. I consider years six to eight to be the "tenure window" to account for people who go up for tenure early or late (because of a leave of absence, for example). I assume that an individual is denied tenure if he moves to a university ranked five positions below the initial institution, or to industry, after six to eight years. For example, a person who moves from the fifth ranked school to the seventh ranked school after six years is not assumed to have been denied tenure whereas someone who moves from the fifth ranked school to the twelfth ranked school after six years is. As another example, a person who moves five or fewer years after his initial appointment is not assumed to have been denied tenure since he moved before the tenure window starts. The tenure institution is the first institution at which an individual goes up for tenure, so if someone moves from university A to university B after 3 years, the tenure institution is university B. For further details, see Sarsons (2016).

I use the RePEc/IDEAS ranking of economics journals to control for the quality of a person's publications. I take the top 85 journals and give the top journal a score of 85. The lowest quality journal has a score of one. Finally, I include current citations of pre-tenure papers as a control variable. These citations were scraped from Google Scholar.

Table 1 presents summary statistics of the data. Approximately 70 percent of the full sample receive tenure at the first institution they went up for tenure at, but this masks a stark difference between men and women. Only 52 percent of women receive tenure while 75 percent of men do. There is no statistically significant difference in the number of papers that men and women produce. The only statistically significant

TABLE 1—SUMMARY STATISTICS

	Full	Male	Female	<i>p</i> -value
Tenure	0.70 (0.46)	0.75 (0.43)	0.52 (0.50)	0.001
Years to tenure	6.8 (1.9)	6.6 (1.9)	7.1 (1.9)	0.012
Total papers	8.5 (4.0)	8.6 (4.1)	8.1 (3.4)	0.187
Solo-authored	3.0 (2.3)	3.0 (2.3)	3.1 (2.3)	0.837
Coauthored	5.5 (3.7)	5.6 (3.8)	5.1 (3.2)	0.120
Total unique coauthors	4.7 (3.2)	4.8 (3.1)	4.6 (3.3)	0.415
Frac. female coauthors	0.12 (0.18)	0.09 (0.16)	0.21 (0.24)	0.001
<i>Average journal rank</i>				
All publications	45.5 (18.8)	46.0 (17.0)	42.67 (17.3)	0.053
Solo publications	46.4 (23.7)	47.0 (24.0)	45.0 (23.7)	0.421
Coauthored publications	44.9 (20.3)	45.7 (20.4)	42.0 (19.7)	0.076
Observations	568	436	132	

productivity difference is that men tend to publish their coauthored papers in slightly better journals (where 85 is the highest-ranking journal). Men and women have roughly the same number of coauthors (*Total unique coauthors*), but women are twice as likely to coauthor with women than men are.

II. Empirical Results

A. Main Results

Table 1 shows a significant tenure gap between men and women. I now test whether the composition of papers (solo versus coauthored) matters for tenure. Figure 1 plots the relationship between the fraction of one's papers that are solo-authored and the probability of receiving tenure. Specifically, it plots $\hat{\beta}_1$ and $\hat{\beta}_2$ from

$$\begin{aligned}
 (1) \quad T_{ifst} = & \beta_1 \text{FracSolo}_i + \beta_2 (\text{fem}_i \times \text{FracSolo}_i) \\
 & + \beta_3 \text{TotalPapers}_i + \beta_4 \text{fem}_i + \gamma' \mathbf{Z}_i \\
 & + \theta_f + \theta_s + \theta_t + \epsilon_{ifst}.
 \end{aligned}$$

The lines of best fit are plotted after controlling for total papers, the average ranking of coauthored and solo-authored journals the individual publishes in; time to tenure; and school, year, and field fixed effects. For men, the composition of papers does not matter. Conditional on the quality of the paper, a man's tenure rate is approximately 75 percent, regardless of whether he coauthors or solo-authors. This is not the case for women. While women who only write solo-authored papers have comparable tenure rates to men, women who coauthor all of their work have only a 40 percent chance of receiving tenure.

The probability of receiving tenure is not lower for all women who coauthor. In Table 2, I categorize coauthored papers into those written with only men, only women, or a mix of men and women:

$$\begin{aligned}
 (2) \quad T_{ifst} = & \beta_1 S_i + \beta_2 (fem_i \times S_i) + \beta_3 CA_{male}_i \\
 & + \beta_4 (fem \times CA_{male}_i) + \beta_5 CA_{mix}_i \\
 & + \beta_6 (fem \times CA_{mix}_i) + \beta_7 CA_{fem}_i \\
 & + \beta_8 (fem_i \times CA_{fem}_i) + \beta_9 fem_i \\
 & + \gamma' Z_i + \theta_f + \theta_s + \theta_t + \epsilon_{ifst}.
 \end{aligned}$$

Here, S_i is the number of solo-authored papers individual i has; CA_{fem}_i is the number of coauthored papers individual i has in which all of the coauthors are female; CA_{male}_i is the number of papers i has in which all of the coauthors are male and CA_{mix}_i is the number of papers i has in which the coauthors consist of men and women.

The estimated coefficients on the interaction terms show that women receive almost no marginal benefit from producing a coauthored paper with a man but receive some benefit when there is either a mix of male and female coauthors or when they only coauthor with women. These estimates are imprecise due to sample size, but the point estimates suggest that an additional paper that a woman writes with a mix of men and women improves tenure chances by 4.5 percentage points and an additional paper with other women improves tenure probability by 8 percentage points. The estimates are robust to controlling for the total number of coauthors on a paper, as well as paper quality measures, field, tenure institution, and tenure year fixed effects.

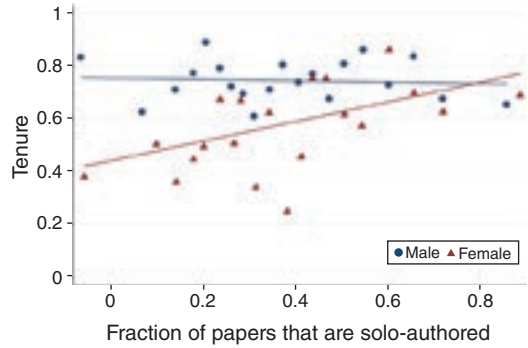


FIGURE 1. RELATIONSHIP BETWEEN PAPER TYPE AND TENURE

Notes: This figure is a binned scatterplot of the correlation between tenure and the fraction of an individual's papers that are solo-authored, split by gender. Both variables are residualized on the following controls before plotting: number of years it took to go up for tenure, average journal rank for solo publications, average journal rank for coauthored publications, total citations, and tenure school, tenure year, and field fixed effects. The line of best fit using OLS is shown separately for men and women. The lines of best fit are estimated using the full sample ($N = 552$) and have slopes of $\beta = 0.41$ (s.e. = 0.17) for women and $\beta = -0.05$ (s.e. = 0.08) for men. The y variable is a binary variable indicating whether an individual received tenure. Each dot represents the mean of approximately 26 observations along both dimensions.

B. Clear Signals: Testing Against Other Authorship Conventions

If the uncertainty contained in a "coauthored" signal is contributing to the gender gap, clearly stating an individual's role on a project should alleviate this problem. I look at sociology to test whether this is true. In sociology, authors are listed by order of contribution, which removes uncertainty over each author's contribution.

Of course it should be noted that sociology is not the perfect comparison as many other factors differ across the two disciplines, including the fraction of women in each discipline. Still, the results shed light on whether the uncertainty contained in coauthored papers in economics contributes to a promotion gap.

The sociology sample consists of randomly sampled faculty at the top 20 sociology PhD-granting schools in the United States. There are 250 sociologists in the sample and 40 percent are female. I test whether men and women are treated differently when they

TABLE 2—RELATIONSHIP BETWEEN PAPERS AND TENURE

	Dep var: Tenure (1)	
	× Female	
Solo-authored pubs	0.068 (0.009)	0.005 (0.015)
Pubs with only male CAs	0.072 (0.012)	−0.071 (0.019)
Pubs with male and female CAs	0.096 (0.033)	−0.051 (0.037)
Pubs with only female CAs	0.069 (0.016)	0.012 (0.027)
Total coauthors	−0.001 (0.006)	
School fixed effects	Yes	
Tenure year fixed effects	Yes	
Field fixed effects	Yes	
Observations	559	

Notes: This table presents the results of one regression where the interaction terms are displayed in the right-hand column. The regression controls for average journal quality and log citations. The equation is estimated using a probit model. The marginal probabilities calculated at the mean are displayed. Standard errors, reported in parentheses, are clustered by tenure institution.

coauthor papers in Table 3. I estimate equation 1 but modify it slightly to include the fraction of a researcher's coauthored papers on which she is the first author.

Being first author on papers is strongly correlated with tenure for both men and women. It is associated with a roughly 4 percent increase in tenure probability, regardless of gender. Importantly, women are not penalized for coauthoring. Results are somewhat noisy due to the small sample, but the coefficient on the female/total-coauthored papers interaction term is close to zero and insignificant.

III. Conclusion

While the results presented in this paper are correlations, they provide suggestive evidence that gender plays a role in promotion decisions. The probability that a woman receives tenure is significantly lower when she coauthors unless she works with other women. Men, on the other hand, have an equal probability of receiving tenure when they coauthor and solo author, conditional on the quality of the paper. Being

TABLE 3—NONALPHABETICAL ORDERING: SOCIOLOGY

	(1)	
	× Female	
Fraction first author	0.403 (0.043)	−0.042 (0.172)
Solo papers	0.000 (0.006)	0.007 (0.011)
Total coauthored	0.009 (0.007)	0.001 (0.015)
School fixed effects	Yes	Yes
Tenure year fixed effects	Yes	Yes
Observations	237	209

Notes: This table shows the relationship between paper types and tenure for the sociology sample. The dependent variable is a binary variable indicating whether an individual received tenure six to eight years after being hired. Fraction first author is the fraction of papers an individual has on which they were first author by the time he or she went up for tenure. All regressions control for the number of books published, time to tenure, and include tenure institution and tenure year fixed effects. The equation is estimated using a probit model. The marginal probabilities calculated at the mean are displayed. Standard errors, reported in parentheses, are clustered by tenure institution.

aware of this phenomenon is important as many occupations today require group work. The tech industry, for example, prides itself on collaboration. In such male-dominated fields, however, group work could result in fewer women moving up the career ladder if credit is not properly attributed.

REFERENCES

- Antecol, Heather, Kelly Bedard, and Jenna Stearns. 2016. "Equal but Inequitable: Who Benefits from Gender-Neutral Tenure Clock Stopping Policies?" IZA Discussion Paper 9904.
- Ceci, Stephen J., Donna K. Ginther, Shulamit Kahn, and Wendy M. Williams. 2014. "Women in Academic Science: A Changing Landscape." *Psychological Science in the Public Interest* 15 (3): 75–141.
- Ginther, Donna K., and Shulamit Kahn. 2004. "Women in Economics: Moving Up or Falling Off the Academic Career Ladder?" *Journal of Economic Perspectives* 18 (3): 193–214.

- Lazear, Edward, and Kathryn Shaw.** 2007. "Personnel Economics: The Economist's View of Human Resources." *Journal of Economic Perspectives* 21 (4): 91–114.
- Niederle, Muriel, and Lise Vesterlund.** 2007. "Do Women Shy Away from Competition? Do Men Compete Too Much?" *Quarterly Journal of Economics* 122 (3): 1067–101.
- Sarsons, Heather.** 2016. "Recognition for Group Work." <http://scholar.harvard.edu/sarsons/publications/note-gender-differences-recognition-group-work> (accessed December 2016).

This article has been cited by:

1. Dachin N. Frances, Connor R. Fitzpatrick, Janet Koprivnikar, Shannon J. McCauley. 2020. Effects of Inferred Gender on Patterns of Co-Authorship in Ecology and Evolutionary Biology Publications. *The Bulletin of the Ecological Society of America* **69**. . [[Crossref](#)]
2. Shelagh K. Mooney. 2020. Gender research in hospitality and tourism management: time to change the guard. *International Journal of Contemporary Hospitality Management* **32**:5, 1861-1879. [[Crossref](#)]
3. Pallab Ghosh, Zexuan Liu. 2020. Coauthorship and the gender gap in top economics journal publications. *Applied Economics Letters* **27**:7, 580-590. [[Crossref](#)]
4. Jessica C Lai. 2020. The Role of Patents as a Gendered Chameleon. *Social & Legal Studies* **364**, 096466392091623. [[Crossref](#)]
5. Samara Klar, Yanna Krupnikov, John Barry Ryan, Kathleen Searles, Yotam Shmargad. 2020. Using social media to promote academic research: Identifying the benefits of twitter for sharing academic work. *PLOS ONE* **15**:4, e0229446. [[Crossref](#)]
6. Marisa Turesky, Mildred E. Warner. 2020. Gender Dynamics in the Planning Workplace. *Journal of the American Planning Association* **86**:2, 157-170. [[Crossref](#)]
7. Lynda Powell. 2020. GENDER BIAS IN LEGISLATIVE STUDIES?. *PS: Political Science & Politics* **53**:2, 302-304. [[Crossref](#)]
8. . The Production of Knowledge **20**, . [[Crossref](#)]
9. Lamar Pierce, Laura W. Wang, Dennis J. Zhang. 2020. Peer Bargaining and Productivity in Teams: Gender and the Inequitable Division of Pay. *Manufacturing & Service Operations Management* . [[Crossref](#)]
10. Joseph Saraceno. 2020. Disparities in a Flagship Political Science Journal? Analyzing Publication Patterns in the JOP, 1939-2019. *The Journal of Politics* . [[Crossref](#)]
11. Guy Madison, Pontus Fahlman. 2020. Sex differences in the number of scientific publications and citations when attaining the rank of professor in Sweden. *Studies in Higher Education* **15**, 1-22. [[Crossref](#)]
12. Andrés García-Suaza, Jesús Otero, Rainer Winkelmann. 2020. Predicting early career productivity of PhD economists: Does advisor-match matter?. *Scientometrics* **122**:1, 429-449. [[Crossref](#)]
13. Elise Tak, Shelley J Correll, Sarah A Soule. 2019. Gender Inequality in Product Markets: When and How Status Beliefs Transfer to Products. *Social Forces* **98**:2, 548-577. [[Crossref](#)]
14. Alexandra Graddy-Reed, Lauren Lanahan, Jonathan Eyer. 2019. Gender discrepancies in publication productivity of high-performing life science graduate students. *Research Policy* **48**:9, 103838. [[Crossref](#)]
15. Alan Benson, Danielle Li, Kelly Shue. 2019. Promotions and the Peter Principle*. *The Quarterly Journal of Economics* **134**:4, 2085-2134. [[Crossref](#)]
16. Jill J. McCluskey. 2019. Why diversity and expectations matter. *Agricultural Economics* **50**:S1, 107-111. [[Crossref](#)]
17. Javier Gómez-Ferri, Gregorio González-Alcaide, Ramón LLopis-Goig. 2019. Measuring dissatisfaction with coauthorship: An empirical approach based on the researchers' perception. *Journal of Informetrics* **13**:4, 100980. [[Crossref](#)]
18. Joyce J. Chen, Daniel Crown. 2019. The Gender Pay Gap in Academia: Evidence from the Ohio State University. *American Journal of Agricultural Economics* **101**:5, 1337-1352. [[Crossref](#)]
19. Rachel L. Roper. 2019. Does Gender Bias Still Affect Women in Science?. *Microbiology and Molecular Biology Reviews* **83**:3. . [[Crossref](#)]

20. Isis H. Settles, Sheila T. Brassel, Patricia A. Soranno, Kendra Spence Cheruvilil, Georgina M. Montgomery, Kevin C. Elliott. 2019. Team climate mediates the effect of diversity on environmental science team satisfaction and data sharing. *PLOS ONE* 14:7, e0219196. [[Crossref](#)]
21. Roger E. Backhouse, Béatrice Cherrier. 2019. Paul Samuelson, gender bias and discrimination. *The European Journal of the History of Economic Thought* 62, 1-28. [[Crossref](#)]
22. Christine Wiedman. 2019. Rewarding Collaborative Research: Role Congruity Bias and the Gender Pay Gap in Academe. *Journal of Business Ethics* 27. . [[Crossref](#)]
23. Lauren A. Rivera, András Tilcsik. 2019. Scaling Down Inequality: Rating Scales, Gender Bias, and the Architecture of Evaluation. *American Sociological Review* 84:2, 248-274. [[Crossref](#)]
24. Leah Boustan, Andrew Langan. 2019. Variation in Women's Success across PhD Programs in Economics. *Journal of Economic Perspectives* 33:1, 23-42. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
25. Kasey Buckles. 2019. Fixing the Leaky Pipeline: Strategies for Making Economics Work for Women at Every Stage. *Journal of Economic Perspectives* 33:1, 43-60. [[Abstract](#)] [[View PDF article](#)] [[PDF with links](#)]
26. Nichole A Broderick, Arturo Casadevall. 2019. Gender inequalities among authors who contributed equally. *eLife* 8. . [[Crossref](#)]
27. Nour S. Kteily, Matthew D. Rocklage, Kaylene McClanahan, Arnold K. Ho. 2019. Political ideology shapes the amplification of the accomplishments of disadvantaged vs. advantaged group members. *Proceedings of the National Academy of Sciences* 116:5, 1559-1568. [[Crossref](#)]
28. Keyvan Vakili, Florenta Teodoridis, Michaël A. Bikard. 2019. The Collaboration Credit Premium and the Case of Detrimental Collaborations: Evidence from Economics. *SSRN Electronic Journal* . [[Crossref](#)]
29. Laura Hospido, Carlos Sanz. 2019. Gender Gaps in the Evaluation of Research: Evidence from Submissions to Economics Conferences. *SSRN Electronic Journal* . [[Crossref](#)]
30. Sarah Jacobson. 2018. Inclusion Strengthens and Enriches the NAREA Community. *Agricultural and Resource Economics Review* 47:3, 415-418. [[Crossref](#)]
31. Kirstin Early, Jessica Hammer, Megan Kelly Hofmann, Jennifer A. Rode, Anna Wong, Jennifer Mankoff. 2018. Understanding Gender Equity in Author Order Assignment. *Proceedings of the ACM on Human-Computer Interaction* 2:CSCW, 1-21. [[Crossref](#)]
32. Carissa L. Tudor, Deborah J. Yashar. 2018. Gender and the Editorial Process: World Politics , 2007–2017. *PS: Political Science & Politics* 51:4, 870-880. [[Crossref](#)]
33. Ralph Hansmann, Dagmar Schröter. 2018. Equal Opportunities in Academic Careers? How Mid-Career Scientists at ETH Zurich Evaluate the Impact of Their Gender and Age. *Sustainability* 10:9, 3343. [[Crossref](#)]
34. Ana Maria Takahashi, Shingo Takahashi, Thomas N. Maloney. 2018. Gender gaps in STEM in Japanese academia: The impact of research productivity, outside offers, and home life on pay. *The Social Science Journal* 55:3, 245-272. [[Crossref](#)]
35. Madeline E. Heilman, Suzette Caleo. 2018. Combatting gender discrimination: A lack of fit framework. *Group Processes & Intergroup Relations* 21:5, 725-744. [[Crossref](#)]
36. Hannah Rubin, Cailin O'Connor. 2018. Discrimination and Collaboration in Science. *Philosophy of Science* 85:3, 380-402. [[Crossref](#)]
37. Marc J. Lerchenmueller, Olav Sorenson. 2018. The gender gap in early career transitions in the life sciences. *Research Policy* 47:6, 1007-1017. [[Crossref](#)]
38. Derek C. Jones. 2018. The economics of Participation and Employee Ownership (PEO): an assessment. *Journal of Participation and Employee Ownership* 1:1, 4-37. [[Crossref](#)]

39. William P. Metheny, Mayuri Jagadish, R. Eric Heidel. 2018. A 15-Year Study of Trends in Authorship by Gender in Two U.S. Obstetrics and Gynecology Journals. *Obstetrics & Gynecology* **131**:4, 696-699. [[Crossref](#)]
40. Lukas Kuld, John O'Hagan. 2018. Rise of multi-authored papers in economics: Demise of the 'lone star' and why?. *Scientometrics* **114**:3, 1207-1225. [[Crossref](#)]
41. David Ong, Ho Fai Chan, Benno Torgler, Yu (Alan) Yang. 2018. Collaboration incentives: Endogenous selection into single and coauthorships by surname initial in economics and management. *Journal of Economic Behavior & Organization* **147**, 41-57. [[Crossref](#)]
42. Karla Hoff, James Walsh. 2018. The Whys of Social Exclusion: Insights from Behavioral Economics. *The World Bank Research Observer* **33**:1, 1-33. [[Crossref](#)]
43. Lamar Pierce, Laura W. Wang, Dennis Zhang. 2018. Peer Bargaining and Productivity in Teams: Gender and the Inequitable Division of Pay. *SSRN Electronic Journal* . [[Crossref](#)]
44. Cleo Chassonnery-Zaagouche, Catherine Sophia Herfeld, Erich Pinzzn-Fuchs. 2018. New Scope, New Sources, New Methods? An Essay on Contemporary Scholarship in History of Economic Thought Journals, 2016-2017. *SSRN Electronic Journal* . [[Crossref](#)]
45. Roger Backhouse, Beatrice Cherrier. 2018. 'I Occasionally Learn Something': Paul Samuelson, Gender Bias and Discrimination Before 1973. *SSRN Electronic Journal* . [[Crossref](#)]
46. James J. Heckman, Sidharth Moktan. 2018. Publishing and Promotion in Economics: The Tyranny of the Top Five. *SSRN Electronic Journal* . [[Crossref](#)]
47. Christine I. Wiedman. 2018. Rewarding Collaborative Research: An Analysis of Gender and Academic Accounting Faculty Salaries. *SSRN Electronic Journal* . [[Crossref](#)]
48. Andreas Born, Eva Ranehill, Anna Sandberg. 2018. A Man's World? The Impact of a Male Dominated Environment on Female Leadership. *SSRN Electronic Journal* . [[Crossref](#)]
49. Stephanie E. Hampton, Stephanie G. Labou. 2017. Careers in ecology: a fine-scale investigation of national data from the U.S. Survey of Doctorate Recipients. *Ecosphere* **8**:12, e02031. [[Crossref](#)]
50. Ana María Munar. 2017. To be a feminist in (tourism) academia. *Anatolia* **28**:4, 514-529. [[Crossref](#)]
51. 2017. Energizing equality. *Nature Energy* **2**:10, 759-759. [[Crossref](#)]
52. Co-Pierre Georg, Michael Ernst Rose. 2015. Mirror, Mirror, on the Wall, Who Is the Most Central of Them All?. *SSRN Electronic Journal* . [[Crossref](#)]